Type 2: Application Project

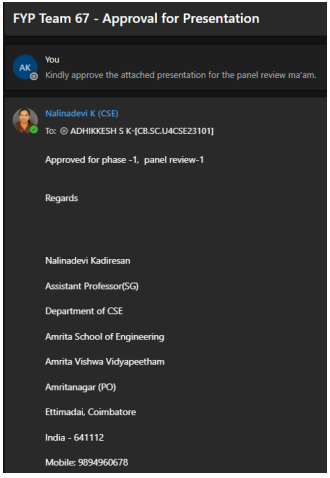
Multimodal Human Activity and Sign Language Recognition for Inclusive Smart Homes

23CSE399 – Project Phase 2
Guide Review 1 | Panel Review 1

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Guide Approval



Guide approval

1. Motivation

- **Smart Homes:** Rapid adoption — **82.1% projected increase in smarthome systems by 2026** *(Statista)*
- **Accessibility:** **1.5 billion** people with hearing loss → **2.5 billion by 2050** *(WHO)*
- **Sign-Language Recognition:** Since **1991** — predominantly **camera/image- or wearable sensor-based recognition**
- **Human Activity Recognition:** Predominantly **single-modality sensing** → limited information from one sensing source
- **Privacy:** **87% of reviewed studies** reported privacy concerns with smart-home surveillance *(JMIR Scoping Review)*

Most existing systems rely on **single-modality sensing**, limiting recognition of **activities, sign language, interactions, and intent**. Along with **camera privacy concerns** and the need for **accessible smart homes**, this motivates a **multimodal, privacy-preserving approach**.

2. Introduction

- **Context Awareness:** *What is happening?* $\rightarrow$ Person sitting, walking, cooking
- **Intent Awareness:** *Why is it happening?* Opening a door $\rightarrow$ To leave, to let a guest in, to get fresh air
- **Context $\neq$ Intent:** The same activity can represent **different intentions** depending on the situation.
- **Real-World Complexity:** **Multiple people + simultaneous activities + interactions** make intent harder to infer.
- **Beyond Activity Recognition:** Understanding **who is doing what, with whom, and why** enables more meaningful automation.
- **Our Goal:** Move from **activity recognition $\rightarrow$ context understanding $\rightarrow$ intent-aware smart-home decisions**.

3. Literature Survey: Human Activity Recognition (HAR)

Paper / Year / Author	Problem Addressed	Dataset	Technique	Result & Limitation
LAHAR (2024) Chen et al. [1]	Differentiating multiple people and recognizing concurrent activities.	Ambient *(Sensor Logs)*	Converts sensor events into text and uses an LLM to reason about activities.	**Result:** Strong performance on ARAS multi-person data. **Limit:** Lacks fine hand/finger motion detection, creating context blindness.
Multi-Skel (2024) Li et al. [2]	Reducing the complexity of multi-person tracking while preserving accuracy.	Video / Skel *(Vision)*	Detects and tracks skeleton keypoints for several people from video.	**Result:** 98.0% accuracy with improved tracking on MOT16. **Limit:** Body occlusion hides keypoints, reducing reliable recognition.
PA-CSI (2025) Quy et al. [3]	Combining WiFi amplitude and phase for robust activity recognition.	WiFi CSI *(RF Signals)*	Combines WiFi amplitude and phase through attention-based fusion.	**Result:** 99.9% in-room and 98.0% unseen room accuracy. **Limit:** High accuracy, but motion from multiple people overlaps in RF signals.
Interact-Obj (2025) Wen et al. [4]	Using interacted objects to improve recognition beyond skeletal features.	RGB-D *(Skel+Obj)*	Adds the interacted object as an extra node in the skeleton graph.	**Result:** 96.7% cross-subject and 99.2% cross-view on NTU-60. **Limit:** Vision-based system still struggles with simultaneous actions.
AttCLHAR (2024) Ref. [5]	Suppressing noisy skeletal features to improve activity recognition.	Wearable *(IMU Sensors)*	Uses self-attention over wearable accelerometer and gyroscope signals.	**Result:** 98.1% on WISDM and 96.0% on USC-HAD. **Limit:** Wearable signals provide little context about surrounding people.
OAHR (2024) Ref. [6]	Maintaining recognition when some skeleton joints are hidden or missing.	Skeleton *(Occluded)*	Represents the skeleton as a 3D volume to handle missing joints.	**Result:** 97.1% on NTU RGB+D 60 with occlusion robustness. **Limit:** Vision-based; does not provide privacy-safe multi-user separation.

Research Gap: Single-modality systems suffer from context blindness, while vision-based frameworks can degrade under occlusion or multi-user overlap. The use of LLMs, object-aware skeletons, and attention-based learning shows progress, but a **robust multimodal fusion framework** is still needed for complex parallel activities.

4. Literature Survey: Sign Language Recognition

Paper / Year / Author	Problem Addressed	Dataset	Technique	Result & Limitation
SignFi (2018) Ma et al. [7]	Camera-free sign gesture recognition in home settings.	SignFi *(WiFi CSI)*	CNN tracking hand/finger motion from WiFi signal variations.	**Result:** 98.9% (known), 86.6% (unseen). **Limit:** Nearby motion corrupts target sign signal.
CorrNet (2023) Hu et al. [8]	Tracking hand and face movements in continuous video.	PHOENIX *(RGB Video)*	Correlates hand and face regions without explicit pose estimation.	**Result:** 19.4% Word Error Rate. **Limit:** Vulnerable to occlusion and privacy issues.
LLMs SLT (2024) Gong et al. [9]	Connecting visual sign tokens to LLMs for translation.	PHOENIX *(RGB Video)*	Converts sign video into discrete tokens for LLM translation.	**Result:** BLEU-1 45.2 vs 33.2 baseline. **Limit:** Camera-based; fails with background motion.
Wi-SL (2023) Wang et al. [10]	Extracting fine hand motion from noisy WiFi CSI.	Custom *(WiFi CSI)*	CNN processing with nearest-neighbor matching across rooms.	**Result:** 98.9% gesture accuracy. **Limit:** Single-signer focus; overlapping degrades performance.
WatchSign (2023) Liu et al. [11]	Daily sign language recognition using wearable devices.	Custom *(Wearable IMU)*	Smartwatch and smart ring sensors capture wrist/finger motion.	**Result:** 94.2% accuracy on 100 ASL words. **Limit:** Cannot detect ambient multi-person activities.
ASL Trans. (2024) Gu et al. [12]	Continuous sentence-level ASL translation without cameras.	Custom *(Wearable IMU)*	Wearable inertial motion capture decoded via Bi-LSTM.	**Result:** 99.0% word, 97.3% sentence accuracy. **Limit:** Full-body suit is too intrusive for daily home use.
RF-ASLR (2026) Nosheen et al. [13]	Privacy-preserving sign language recognition via RF sensing.	Custom *(Wireless CSI)*	Software-defined radio CSI temporal patterns decoded via LSTM.	**Result:** 98.0% accuracy across 20 ASL signs. **Limit:** Tested only in controlled, static, single-user setups.

Research Gap: Existing single-modality models (WiFi CSI, Vision, IMU) suffer performance drops amid multi-person background activity. A **multimodal fusion framework** combining WiFi CSI with complementary contactless sensing is required to isolate sign language intent from environmental interference while maintaining privacy.

5. Problem Statement

Challenge:

How can smart homes reliably recognize intentional communicative gestures without relying on intrusive camera-based sensing?

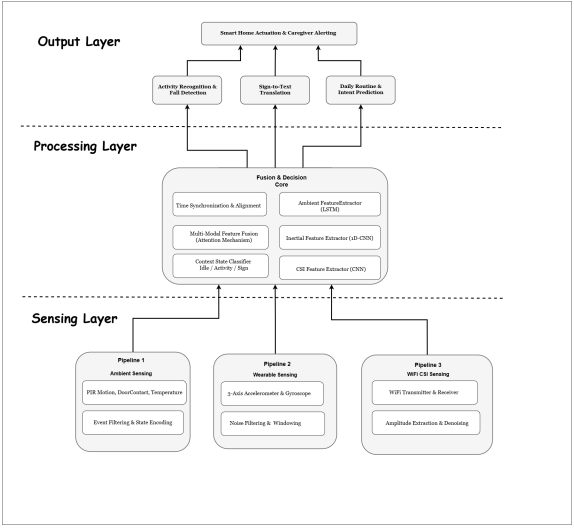
Problem Statement:

To build a privacy-preserving multimodal sensing system that distinguishes routine activities from intentional non-verbal communication, understands user intent through contextual information, and enables appropriate smart-home control without relying on intrusive camera-based sensing.

6. Research Objectives

- **RO1:** To design a privacy-preserving, single-person, single-modality data acquisition methodology, and to develop classification models that distinguish daily *Activities* from *Sign Gestures*.
- **RO2:** To extend this methodology to a single-person, multi-modality setting, and develop classification models that accurately recognize complex, interleaved activities performed by a single individual.
- **RO3:** Extension to multi-person, multi-modality sign and activity recognition system capable of accurately identifying parallel and concurrent activities across multiple individuals within a shared environment.
- **RO4:** To develop multimodal fusion strategies that establish comprehensive context and intent awareness, seamlessly translating deliberate non-verbal communication into privacy-preserving smart home controls.

8. System Architecture



9. System Evolution: From Passive Capture to Controlled CSI

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

SETUP 1

Phone Hotspot

↓

Laptop WiFi Traffic

↓

Single USRP
Passive CSI Capture

- Passive observation
- Traffic dependent
  - Limited channel control

⇒

SETUP 2

USRP N210 TX

↓

Wireless Channel

↓

USRP N210 RX

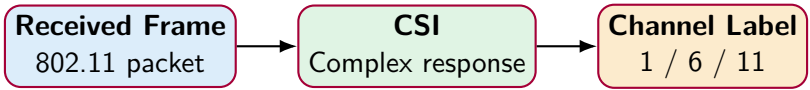
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CSI Extraction

- Controlled transmission
- Dedicated TX/RX
  - Channel hopping

10. CSI Data: From Channel Response to HAR Input

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.



CSI Representation

Complex channel response measured across the OFDM subcarriers.

$$H_k = |H_k| e^{j \angle H_k}$$

Amplitude and phase provide the sensing information.

HAR Input

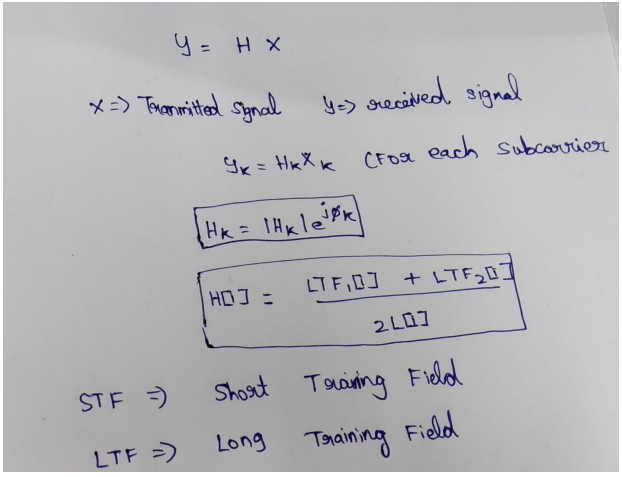
CSI samples are arranged over time to capture motion-induced channel variations.

Input: temporal CSI variations
Output target: activity / sign class

Key result: Each CSI sample is associated with the active WiFi channel, enabling channel-aware HAR analysis.

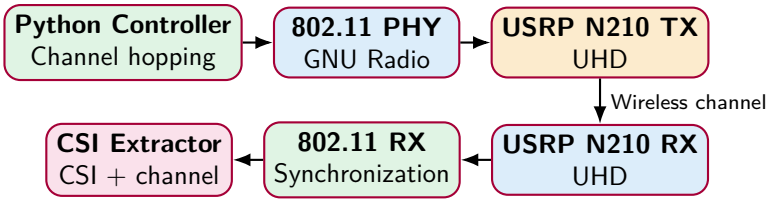
11. CSI Mathematical Model

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.



12. CSI Processing and Software Architecture

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.



TX / Channel Control

- ✓ Python-controlled frequency hopping
- ✓ GNU Radio 802.11 PHY
- ✓ UHD control of USRP N210 TX
- ✓ Channels 1, 6 and 11

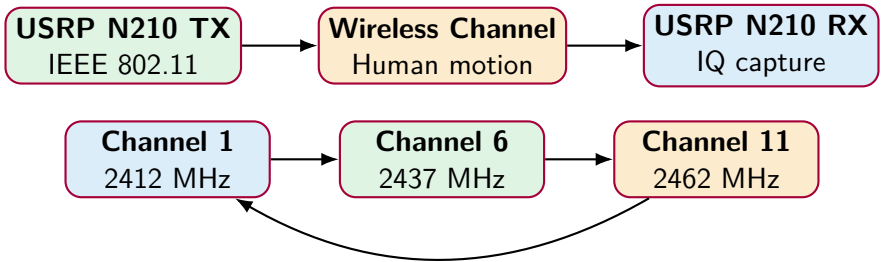
RX / CSI Processing

- ✓ USRP N210 IQ reception
- ✓ 802.11 synchronization
- ✓ Frame equalization
- ✓ Custom CSI extraction

Current output: CSI measurements associated with the active channel.

13. Current Implementation: Two-USRP CSI Setup

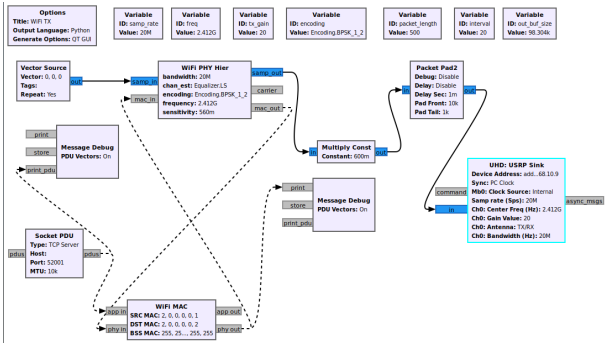
RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.



- Controlled IEEE 802.11 transmission
- Dedicated SDR transmitter and receiver
- Current collection: fixed Channel 1 (2412 MHz)
- Hopping retained as an optional hardware capability

14. CSI Transmission Pipeline

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

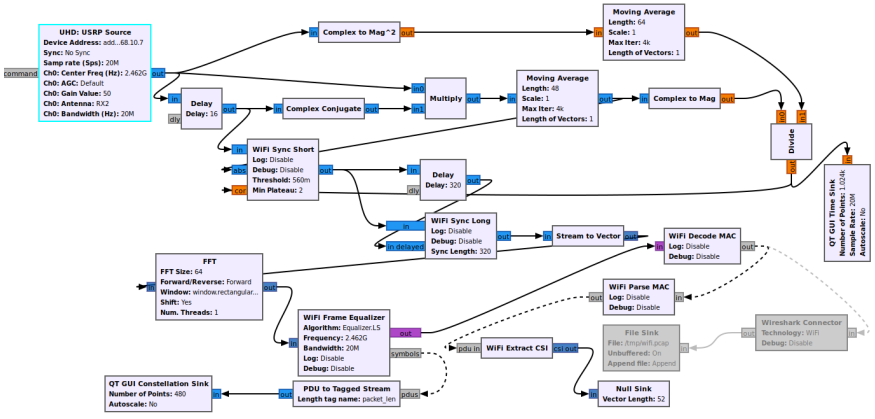


Key result:

Network payloads are encoded into 802.11 WiFi frames and transmitted via USRP to enable continuous channel measurement.

15. CSI Acquisition Pipeline

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.



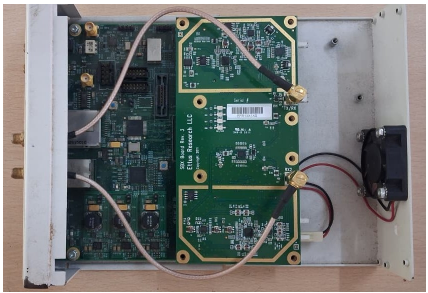
Key result: Received WiFi frames are converted into CSI measurements for subsequent HAR analysis.

16. USRP N210 Hardware

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.



Front panel — RF, MIMO expansion, Gigabit Ethernet, PPS/REF sync



Internal SBX daughterboard — TX/RX and RX2 RF front-end

Ettus Research USRP N210 ▪ **SBX daughterboard** ▪ Gigabit Ethernet host interface ▪ two units used as dedicated TX / RX pair

17. Active vs. Passive CSI Sensing

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Passive Sensing



How it works:

- Phone and laptop are the real TX/RX pair
- USRP transmits nothing; it only listens
- CSI is overheard from existing traffic
- Traffic is uncontrolled, app-dependent

Active Sensing



How it works:

- One USRP actively transmits 802.11 frames
- The second USRP passively receives & extracts CSI
- Channel, rate, and timing are fully controlled

18. Dataset 1: Action vs. No-Action

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Action

- Walking
- Mixed activity

No-Action

- Standing still
- Sitting still

Record Schema

- `Packet`, `Timestamp_us`
- 52 subcarriers × (Re, Im) = **104**
- class encoded in **filename**

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Packet	Timestamp	SC0_Re	SC0_Im	SC1_Re	SC1_Im	SC2_Re	SC2_Im	SC3_Re	SC3_Im	SC4_Re	SC4_Im	SC5_Re	SC5_Im
2	0	1.78E+15	0.014596	0.027721	0.024356	0.025646	0.034323	0.022543	0.041275	0.015439	0.044603	0.005832	0.047344	-0.00202
3	1	1.78E+15	-0.00563	0.028448	0.001935	0.035409	0.013318	0.033606	0.018863	0.036343	0.02623	0.027905	0.032915	0.016866
4	2	1.78E+15	0.027478	-0.0038	0.030876	-0.00799	0.031189	-0.01534	0.02741	-0.02502	0.022354	-0.02825	0.017813	-0.02925
5	3	1.78E+15	0.020966	0.016064	0.026954	0.012451	0.032063	0.004848	0.035008	-0.00027	0.034849	-0.00642	0.030908	-0.00948
6	4	1.78E+15	-0.00324	0.022694	0.0025	0.026688	0.009753	0.027839	0.021471	0.025825	0.026056	0.019209	0.027174	0.014928
7	5	1.78E+15	0.022478	-0.00463	0.021354	-0.0096	0.022032	-0.0125	0.019476	-0.01918	0.013815	-0.02158	0.010971	-0.02115
8	6	1.78E+15	0.020496	0.008144	0.023641	0.000894	0.026138	-0.00709	0.026635	-0.01061	0.022055	-0.01226	0.022125	-0.01868
9	7	1.78E+15	0.01525	0.007041	0.018806	0.002185	0.022146	-0.0018	0.023552	-0.00723	0.019835	-0.01009	0.017884	-0.00765
10	8	1.78E+15	0.009099	0.013109	0.016144	0.009974	0.020433	0.006775	0.023476	0.008103	0.022094	-0.00263	0.019417	-0.00508
11	9	1.78E+15	0.01142	0.01025	0.015761	0.007058	0.0187	0.004601	0.019417	0.001332	0.017358	-0.00266	0.021619	-0.00375
12	10	1.78E+15	0.000855	0.014645	0.003812	0.01557	0.009065	0.017262	0.014883	0.016865	0.015803	0.007777	0.021316	0.002714
13	11	1.78E+15	0.005905	0.012384	0.012078	0.01263	0.015825	0.011451	0.019703	0.008201	0.020039	0.003965	0.018809	-0.0038
14	12	1.78E+15	0.012617	-0.00471	0.01507	-0.00791	0.014007	-0.01399	0.015202	-0.01319	0.012211	-0.01709	0.011524	-0.01823
15	13	1.78E+15	-0.00389	0.018881	0.000715	0.019526	0.007203	0.020243	0.011928	0.020307	0.014904	0.016904	0.024599	0.012795
16	14	1.78E+15	-0.00758	0.01818	-0.00248	0.020859	0.000595	0.025511	0.007773	0.024733	0.014608	0.024718	0.017095	0.017285
17	15	1.78E+15	-0.00415	0.019357	0.000992	0.023367	0.007104	0.026898	0.011027	0.027401	0.019093	0.025891	0.025333	0.018149
18	16	1.78E+15	0.009896	0.021986	0.016725	0.021134	0.022581	0.021708	0.024899	0.016724	0.026321	0.012438	0.029049	0.003572
19	17	1.78E+15	-0.01012	0.022036	-0.00447	0.029091	0.004794	0.034717	0.013214	0.032682	0.022707	0.029755	0.023966	0.02466
20	18	1.78E+15	0.026506	0.002497	0.031096	-0.00061	0.032884	-0.00921	0.034859	-0.0151	0.030229	-0.02045	0.030288	-0.02321
21	19	1.78E+15	0.008585	0.027685	0.016292	0.02894	0.025681	0.026403	0.032218	0.026918	0.036051	0.020545	0.035251	0.013513
22	20	1.78E+15	0.01982	0.018497	0.028529	0.01467	0.037146	0.0131	0.039767	0.007677	0.041999	-0.00045	0.039291	-0.00547
23	21	1.78E+15	0.029999	-0.0162	0.025559	-0.02387	0.023588	-0.02887	0.023008	-0.03336	0.020733	-0.03335	0.013736	-0.03434
24	22	1.78E+15	0.0326	0.001576	0.034877	-0.00281	0.038137	-0.00657	0.038322	-0.01256	0.043325	-0.02336	0.040111	-0.03134
25	23	1.78E+15	0.031992	0.004343	0.035494	-0.00287	0.039578	-0.009	0.039674	-0.01731	0.03507	-0.0243	0.031901	-0.02505
26	24	1.78E+15	0.018833	0.025273	0.028016	0.02601	0.03538	0.019709	0.043637	0.018236	0.045615	0.006125	0.047414	-0.00151
27	25	1.78E+15	0.030022	-0.01879	0.030436	-0.02312	0.030962	-0.02973	0.026678	-0.03858	0.019776	-0.0383	0.015847	-0.04202

Raw CSI capture — one row per decoded 802.11 packet

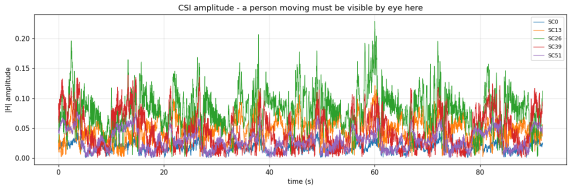
3 participants ▪ **12 takes** ▪ 2 sessions, 2 rooms ▪ same two-USRP setup ▪ 200k records

19. Data Collection: Walking & Standing

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Walking

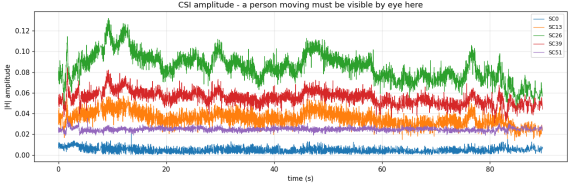
action take · Amrith · Room A

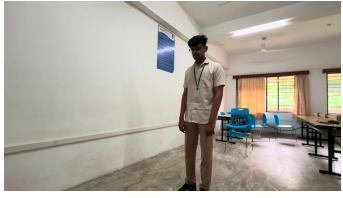




Standing

no-action take · Rishiikesh · Room A



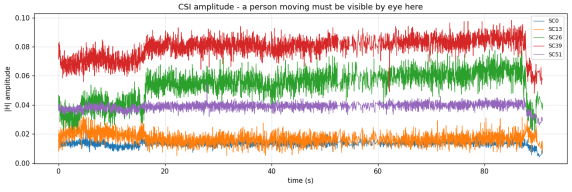


20. Data Collection: Sitting & Mixed Activity

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Sitting

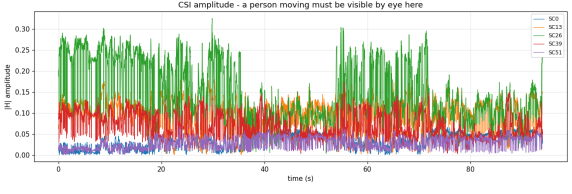
no-action take · Rishiikesh · Room A

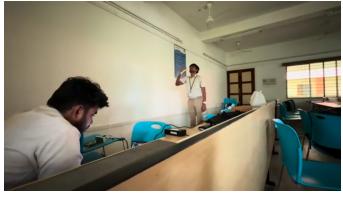




Mixed activity

action take · Gowreesh · Room B





21. Current Dataset: Collection and Coverage

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

5 people **25 kept recordings** **2,050,030 raw rows**

Recording block	Files	CSI rows
Actions: main block	5	844,426
Actions: seated	2	173,297
Signs: standing	5	457,408
Signs: seated	2	127,209
Interleaved stories	5	354,465
Still-person baselines	3	54,094
Empty-room checks	3	39,131
Total collected	**25**	**2,050,030**

Model input: **22 files / 2,010,899 rows**. Recording totals include rest/ready; empty-room files are QC only.



Collection conditions

Two USRP N210s; Channel 1, 2412 MHz.

Same room and antenna positions.

One active participant at a time.

P4/P5 add seated blocks and cooking actions.

22. Current Dataset: Actions and Signs

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

12 action labels

Walk	Sit–stand
Pick up	Light switch
Drink	Answer phone
Eat	Wipe table
TV remote	Fall
Chop food	Stir pot

628 isolated trials. P4/P5 add chop/stir and seven seated action types.

14 sign labels

HELLO	THANKYOU
LAMP	FAN
TELEVISION	COOK
WATER	HOT
HUNGRY	TOILET
CALL	HELP
PAIN	ACCIDENT

490 trials: 350 standing (P1–P5) + 140 seated (P4/P5).

23. Current Dataset: Recording Demonstration

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Standing collection excerpt



Standing participant within the collection area.

Seated collection excerpt



Seated posture and hand movement during recording.

Short excerpts from the recorded collection videos. The label files define the action/sign classes and time intervals.

24. Current Dataset: Interleaved Stories

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.
RO4: Multimodal fusion for context- and intent-aware smart-home control.

Template	Recorded sequence / context
M-SEQ: meal	Walk, light switch, sit–stand, eat/drink; HELLO, HUNGRY, COOK, HOT, WATER, THANKYOU. P4/P5 add chop/stir.
T-SEQ: TV	Walk, sit–stand, TV remote, phone; HELLO, TELEVISION, HOT, LAMP, CALL, TOILET; stand and walk out.
C-SEQ: clean / fall	Walk, wipe, pick up; FAN, HELP; walk and fall; HELP, PAIN, ACCIDENT, CALL on the mattress.
S-SEQ: seated	Wipe, chop, stir, eat, drink; HELLO, HUNGRY, COOK, HOT, WATER, HELP, THANKYOU. P4/P5 only.

Each available template was repeated twice per person: **34 runs / 416 labelled steps** = 208 actions + 208 signs.



Story recording excerpt

Why stories matter

Actions and signs share one recording.

HOT during a meal and HOT while watching TV have different intended responses.

25. Current Dataset: CSI Rows and Annotations

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Raw CSI file: one row per decoded packet

P04_signs_seated_20261016_174235.csv | 107 columns: 3 metadata + 52 × (Re, Im)

Packet	Timestamp_us	Channel	SC0_Re	SC0_Im	SC1_Re	SC1_Im	...
11676	1792152908290345	1	0.000275	-0.000327	0.001651	-0.000113	...
11677	1792152908312995	1	0.000357	-0.000239	0.001407	0.000079	...
11678	1792152908318294	1	0.001165	0.000000	0.001127	-0.000149	...
11679	1792152908386570	1	-0.001413	-0.000463	-0.000872	-0.001368	...

Separate label file: join each packet to its annotated time interval

pid	type	label	posture	t_start_us	t_end_us
P04	sign	ACCIDENT	seated	1792152899283128	1792152903283699
P04	sign	COOK	seated	1792152908284587	1792152912286562
P04	sign	TOILET	seated	1792152917287496	1792152921288077

Selected columns shown. Values are copied from the actual recording and label CSVs; other subcarrier columns are omitted.

Actual CSV excerpt, selected columns. Raw rows contain packet/time/channel and complex CSI; the separate label file supplies participant, type, label, posture and time boundaries. Labels come from the collection cues/key presses; video alignment is not yet verified.

26. Current Dataset: Recording Rows per Class

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Actions

Class	Trials	Raw rows
Answer phone	67	46,741
Chop food	40	25,679
Drink	67	46,554
Eat	67	46,401
Fall	23	16,512
Light switch	48	23,817
Pick up	48	35,094
Sit–stand	48	33,576
Stir pot	40	25,174
TV remote	67	39,134
Walk	46	33,371
Wipe table	67	44,783
Total	**628**	**416,836**

Signs

Class	Trials	Raw rows
ACCIDENT	35	15,864
CALL	35	15,815
COOK	35	15,792
FAN	35	15,956
HELLO	35	16,019
HELP	35	15,782
HOT	35	16,097
HUNGRY	35	16,140
LAMP	35	15,689
PAIN	35	16,051
TELEVISION	35	15,641
THANKYOU	35	15,697
TOILET	35	15,875
WATER	35	16,152
Total	**490**	**222,570**

Raw packets inside fully covered isolated label intervals; rest/ready and story steps excluded. Partial P2 trials excluded. Chop/stir: P4/P5 only.

27. Multi-Modal Extension: Wearable Fall Detection

RO4: Multimodal fusion for context- and intent-aware smart-home control.



Data & Model

- 3-axis accelerometer + gyroscope
- 50 Hz sampling, 2.56 s window
- Class 0: ADL Class 1: falls
- Dual 1D-CNN: 128×3 each stream
- Fused dense layer $\rightarrow$ probability
- Exported to `.tflite`, on-device

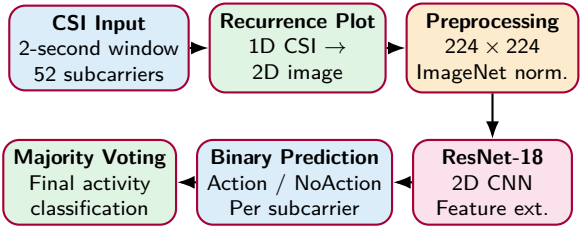
Note — Error Analysis

- **False positive:** placing the phone hard on a table fires the alarm
- **Cause 1:** ADL class held only smooth motion, so the model learnt a shortcut — *any hard impact = fall*
- **Cause 2:** small single-user set — body-on-mattress still looks like phone-on-table
- **V2 fix:** add high-impact *non-fall* ADL data + multi-orientation recordings

Result: ADL 4.8×10^{-4} vs. fall 0.29 — a $\approx$**600**$\times$ relative spike; edge threshold set empirically at **0.20**

28. Earlier Pilot: Binary Action Model

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.



Training:
Adam optimizer + dynamic learning-rate scheduling

Model Description

Model:
Pre-trained ResNet-18 adapted for binary classification.

Input:
2-second temporal CSI windows across 52 subcarriers.

Representation:
CSI signals converted into 2D Recurrence Plots.

Decision:
Predictions are generated for individual subcarriers and combined using majority voting.

29. Earlier Pilot: Action vs. No-Action Results

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Confusion Matrix



Training & Validation Accuracy



Classification Report

Class	Precision	Recall	F1
NoAction	0.93	0.99	0.96
Action	0.99	0.93	0.96
Overall	**0.96**	**0.96**	**0.96**

Real-File Inference

Input: P01_mixed.csv

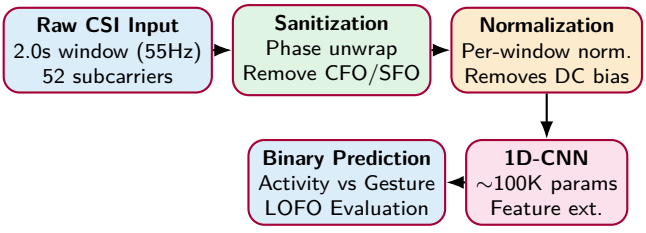
79 windows: 69 Action, 10 No Action

Overall Prediction: Action

Confidence: 87.34%

30. Earlier Pilot: Activity vs. Gesture Model

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.



Training:

30 Epochs + Leave-One-File-Out (LOFO) validation

Model Description

Model:
Lightweight 1D-CNN adapted for raw time-series motion.

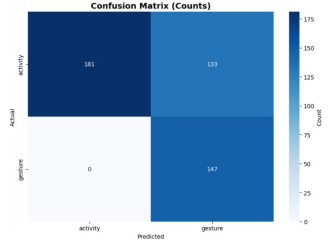
Diagnostic Fix:
Initial 100% accuracy was a false boundary from a mean-amplitude DC bias leak. Fixed via per-window normalization.

Evaluation:
Strict LOFO (Leave-One-File-Out) validation forces the model to learn authentic dynamic motion rather than static room signatures.

31. Earlier Pilot: Activity vs. Gesture Results

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

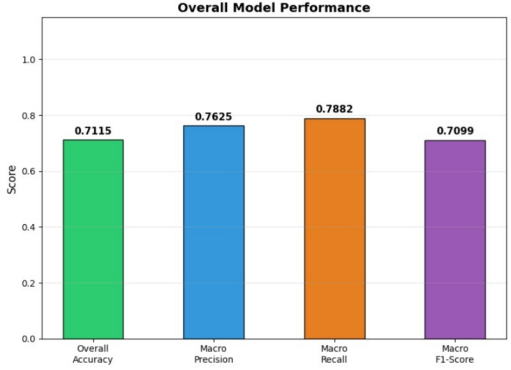
Confusion Matrix



Classification Report

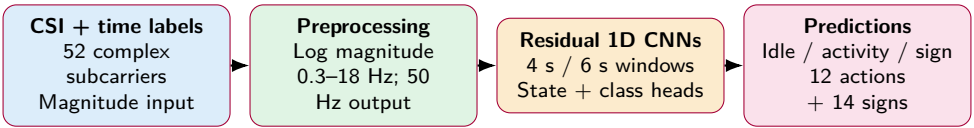
Class	Precision	Recall	F1
Activity	1.00	0.58	0.73
Gesture	0.53	1.00	0.69
Macro Avg	**0.76**	**0.79**	**0.71**

Per-Class & Overall Metrics



32. Current Model: Training and Evaluation

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.



Five held-out-person tests

- Initial fit: 3 people; validation: 1 other person.
- Choose detector/action/sign roles on validation.
- Refit selected networks on 4 people, two seeds.
- Test only on the fifth person; repeat for all five.

All-five deployment model

Two normalized four-second CNNs, one per seed, trained on all five people.

Scales, masks and class order saved.

Kaggle T4 $\times 2$: **41.4 minutes**.

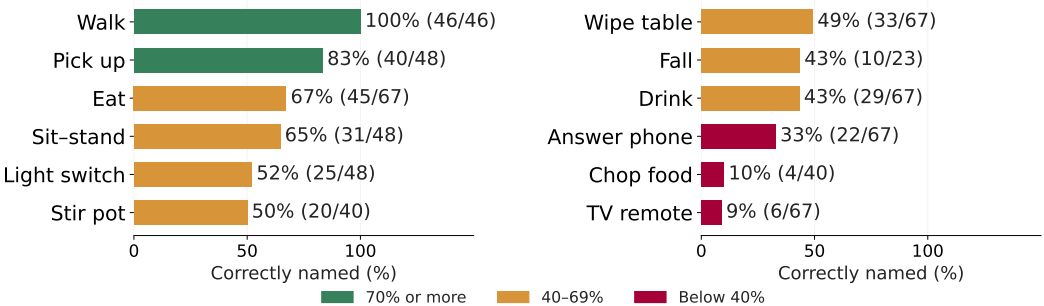
Scores come from held-out folds; the all-five refit has no independent score. Inference is offline. ExtraTrees applies only to isolated action fusion (final weight: zero).

33. Current Results: Action Recognition

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

311/628 correct (49.5% accuracy)

Mean held-out macro F1: **50.2%**



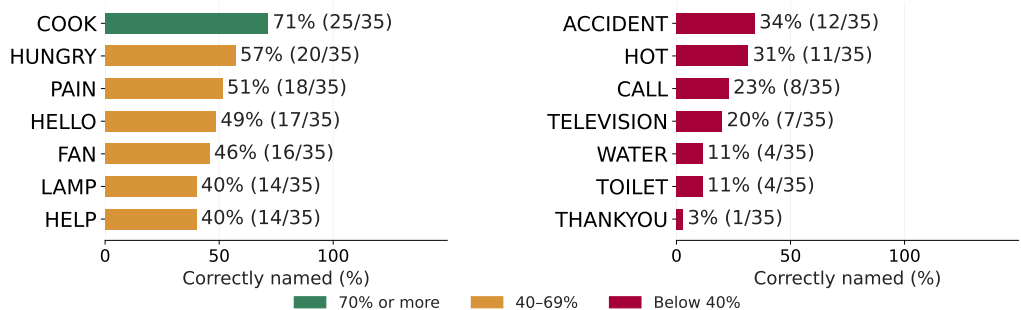
P1–P5 isolated trials, known boundaries/type; bars show per-class recall. **Strongest:** walk/pick up. **Weakest:** TV remote/chop/phone. Chop/stir cover P4/P5 only.

34. Current Results: Sign Recognition

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

171/490 correct (34.9% accuracy)

Mean held-out macro F1: **33.2%**



P1–P5: 350 standing + 140 seated trials; 35 per sign; known boundaries/type. **Stronger:** COOK/HUNGRY/PAIN. **Weakest:** THANKYOU/WATER/TOILET. Isolated seated signs cover P4/P5 only.

35. Current Results: Detection Threshold

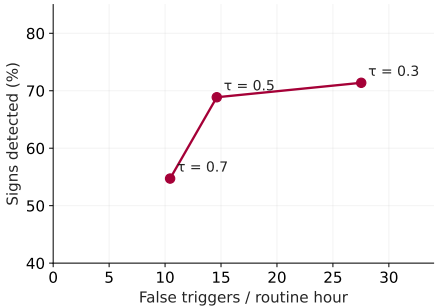
RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.
RO4: Multimodal fusion for context- and intent-aware smart-home control.

Setting	Signs caught	False / hour
$\tau = 0.3$: sensitive	71.4%	27.5
$\tau = 0.5$: middle	68.9%	14.6
$\tau = 0.7$: strict	54.7%	10.5

Raise τ to reduce false triggers; fewer signs are caught.

"Caught" means detected, not correctly named.

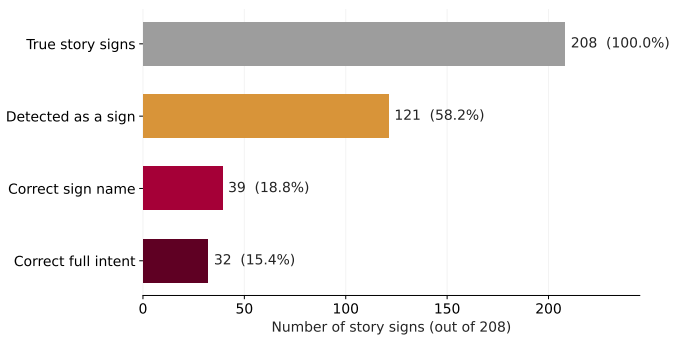
τ is a score cutoff, not calibrated certainty.



Fixed-threshold sweep: P1–P3, isolated + stories, 318 signs; mean over three people. One-second smoothing/minimum run, one-second gap merge and ± 1 s matching. Rates cover routine movement only (about 58 measured minutes), excluding idle/unknown periods.

36. Current Results: Continuous Story Recognition

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.
RO4: Multimodal fusion for context- and intent-aware smart-home control.



Automatic story inference

No true sign boundaries or type are supplied.

58.2% of signs detected.

18.8% correctly named.

15.4% correct full intent.

Continuous identity is the main remaining limitation.

All five people; validation-selected thresholds per fold; one-to-one IoU ≥ 0.10 matching. This differs from the fixed-threshold sweep. There were 50 false triggers during 20.3 minutes of measured story routine activity.

37. MOOC / Course Selection

S.No	Student Name	Chosen MOOC / Course Title	Platform
1	Gowreesh B	Certified Wireless Network Administrator (CWNA)	Udemy
2	Adhikkesh S.K	Basics of Digital Signal Processing for Power Engineers	Udemy
3	Amrith B	Digital Signal Processing 2: Filtering	Coursera
4	Adithya Ajay	Advanced Wireless Communication Systems	
5	Rishiikesh S.K	Signal Processing: Problems Solved in MATLAB and Python	Udemy

38. Limitations

Current Limitations

Bandwidth Limitation

Multi-channel operation requires approximately 70 MHz bandwidth, while our hardware supports only 20 MHz. The sampling rate must also be sufficient to support the required bandwidth.

Continuous Recognition

The current CSI model uses centered 4–6 second windows for offline inference. Continuous sign naming and false-trigger control remain weak; overlapping multi-person recognition has not been evaluated.

Dataset & Generalization

Five people were recorded in one room; isolated seated and cooking examples cover only P4/P5. A fresh session or person is needed to confirm generalization, and cue-based labels still need video verification.

39. References (1/3)

[1] X. Chen *et al.*, "Towards LLM-Powered Ambient Sensor Based Multi-Person Human Activity Recognition (LAHAR)," 2024. [Link]

[2] Li *et al.*, "Multiperson Activity Recognition and Tracking Based on Skeletal Keypoint Detection," 2024. [Link]

[3] Hiremath *et al.*, "Maintenance Required: Updating and Extending Bootstrapped Human Activity Recognition Systems for Smart Homes," 2024. [Link]

[4] H. Wen *et al.*, "Improving Skeleton-based Action Recognition with Interactive Object Information," 2025. [Link]

40. References (2/3)

[5] "Attention-based Contrastive Learning for HAR (AttCLHAR)," 2024. [Link]

[6] "Occlusion-Aware Hypergraph Reasoning for Skeleton Action Recognition (OAHR)," 2024. [Link]

[7] Y. Ma *et al.*, "SignFi: Sign Language Recognition Using WiFi," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 2, no. 1, pp. 1–21, 2018. [Link]

[8] L. Hu *et al.*, "Continuous Sign Language Recognition With Correlation Network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 2529–2539. [Link]

41. References (3/3)

[9] J. Gong *et al.*, "LLMs are Good Sign Language Translators," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 18362–18372. [Link]

[10] H. Wang *et al.*, "Wi-SL: Translating Continuous Sign Language into Text Using Ambient WiFi Signals," 2023. [Link]

[11] Liu *et al.*, "WatchSign: Daily Sign Language Tracking Utilizing Commercial Smartwatch IMU Sensors," 2023.

[12] Y. Gu *et al.*, "End-to-End Continuous Sentence-Level ASL Translation Capturing Whole-Arm Inertial Data," 2024. [Link]